

PART A — THEORETICAL FOUNDATION

1. What is Statistical Distribution?

A statistical distribution describes how the values of a variable are spread or distributed in a dataset. It shows the frequency or probability of different values occurring in the data.

Statistical distributions are used to understand data patterns, analyze behavior, and make predictions. Different types of distributions are used depending on the nature of the data.

For example:

- Transaction success/failure can be modeled using Bernoulli Distribution.
- Number of transactions per day can be modeled using Poisson Distribution.
- Transaction amounts can be modeled using Log-Normal Distribution.

Statistical distributions are widely used in data science, machine learning, finance, business analytics, and research.

2. What is a Q-Q Plot and Why is it Used?

A Q-Q Plot (Quantile-Quantile Plot) is a graphical method used to compare whether a dataset follows a particular theoretical distribution, usually the Normal Distribution.

In a Q-Q Plot:

- If the plotted points follow a straight line, the data is approximately normally distributed.
- If the points deviate from the line, the data is not normally distributed.

Purpose of Q-Q Plot:

- To check normality of data
- To identify skewness
- To detect outliers
- To compare distributions

In this project, the Q-Q Plot was used to test whether transaction amounts follow a normal distribution.

3. Difference Between Discrete and Continuous Distributions

Discrete Distribution

Deals with countable values

Values are separate and distinct

Continuous Distribution

Deals with measurable values

Values can take any value in a range

Discrete Distribution

Example: Number of transactions

Uses distributions like Binomial and Poisson

Discrete data consists of whole numbers, while continuous data can contain decimal values.

Continuous Distribution

Example: Transaction amount

Uses distributions like Normal and Log-Normal

4. What is Bernoulli Distribution?

Bernoulli Distribution is a discrete probability distribution that has only two possible outcomes:

- Success (1)
- Failure (0)

It is used when an experiment has only one trial.

Formula:

$$P(X = x) = p^x(1 - p)^{1-x}$$

Where:

- p = probability of success
- $1 - p$ = probability of failure

Example:

In transaction analysis:

- Successful transaction = 1
- Failed transaction = 0

Bernoulli Distribution helps analyze the probability of transaction success.

5. What is Binomial Distribution?

Binomial Distribution is used to model the number of successes in multiple independent trials.

It is based on:

- Fixed number of trials
- Only two outcomes (success/failure)
- Same probability in every trial

Formula:

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Where:

- n = number of trials
- x = number of successes
- p = probability of success

Example:

Finding the probability of successful transactions in one week.

6. Explain Log-Normal Distribution

A Log-Normal Distribution is a continuous probability distribution where the logarithm of the variable follows a normal distribution.

Characteristics:

- Positively skewed
- Values cannot be negative
- Used for financial and transaction data

Applications:

- Transaction amounts
- Income distribution
- Stock prices

In this project, transaction amounts followed a Log-Normal pattern because higher values occurred less frequently.

7. Explain Power Law Distribution

Power Law Distribution describes situations where small values occur frequently and large values occur rarely.

Characteristics:

- Heavy tail distribution
- Few large observations
- Many small observations

Formula:

$$P(x) \propto x^{-a}$$

Applications:

- Social networks

- Wealth distribution
- Internet traffic
- Financial transactions

In this project, transaction amounts were analyzed using Power Law Distribution to study extreme transaction behavior.

8. What is Box-Cox Transform?

Box-Cox Transform is a statistical transformation technique used to stabilize variance and make data more normally distributed.

Purpose:

- Reduce skewness
- Improve normality
- Improve statistical modeling

Formula:

$$Y(\lambda) = \frac{Y^\lambda - 1}{\lambda}$$

Where:

- λ = transformation parameter

Advantages:

- Improves data symmetry
- Helps in regression analysis
- Reduces variance instability

In this project, Box-Cox Transform was applied to transaction amounts.

9. Explain Poisson Distribution with Example

Poisson Distribution is used to model the number of events occurring within a fixed interval of time or space.

Characteristics:

- Discrete distribution
- Events occur independently
- Average rate remains constant

Formula:

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Where:

- λ = average number of events
- x = number of occurrences

Example:

Number of transactions occurring per day.

In this project, Poisson Distribution modeled daily transaction counts.

10. What is Z-Score Probability?

Z-Score measures how far a value is from the mean in terms of standard deviations.

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X = data value
- μ = mean
- σ = standard deviation

Interpretation:

- Positive Z-score → value above mean
- Negative Z-score → value below mean
- Large Z-score → possible outlier

Applications:

- Probability calculations
- Outlier detection
- Statistical analysis

In this project, Z-score was calculated for transaction amounts.

11. Difference Between PDF and CDF

PDF (Probability Density Function) CDF (Cumulative Distribution Function)

Shows probability density

Shows cumulative probability

Represents likelihood at a point

Represents probability up to a value

Used for continuous distributions

Always increases from 0 to 1

Area under curve = probability

Final value always equals 1